Calc I Lecture 24: More integral properties

We've seen how to define integrals using a limit.
But when is that limit guaranteed to exist?

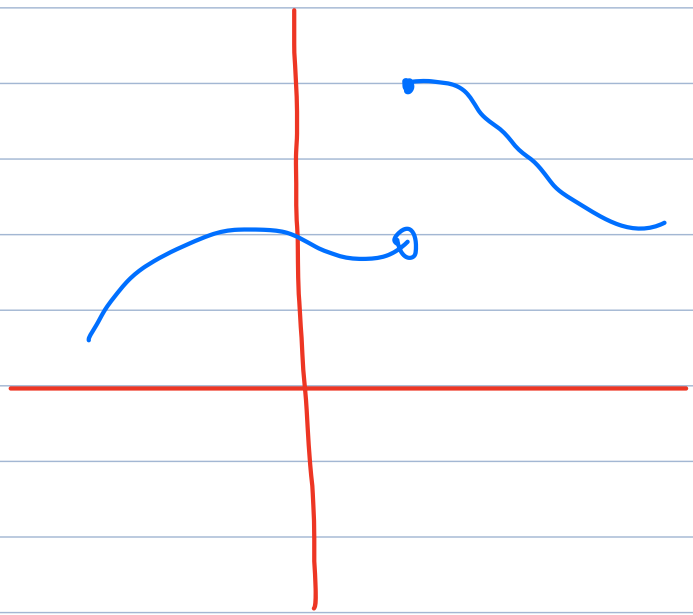
Theorem If f is continuous on $[a,b]$, or if f has only a finite number of jump discontinuities over $[a,b]$ then f is integrable on $[a,b]$. That is, the definite integral

$$\int_a^b f(x)\,dx \text{ exists.}$$

★ note: we only allow jump discontinuities, not infinite

Consequence: practically all functions we use will be integrable unless they have vertical asymptotes over $[a,b]$

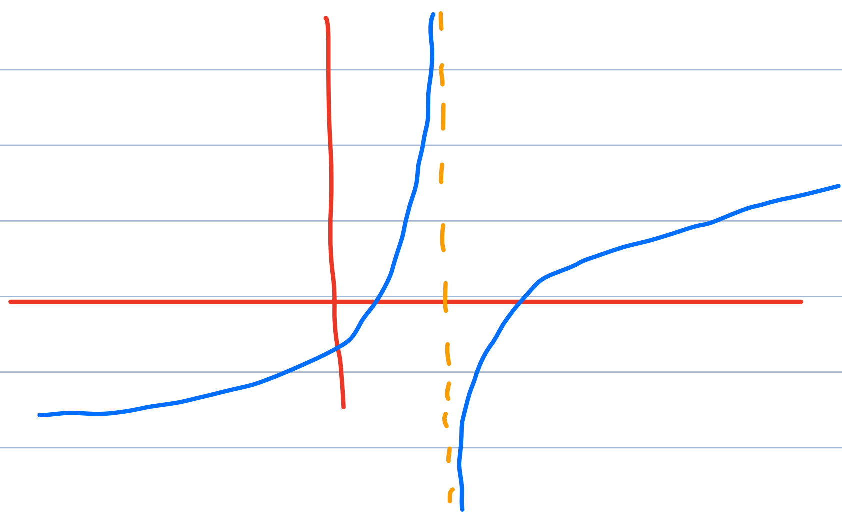
Example of discontinuous integrable function



jump discontinuity

integrable

Example of a non-integrable function



not integrable

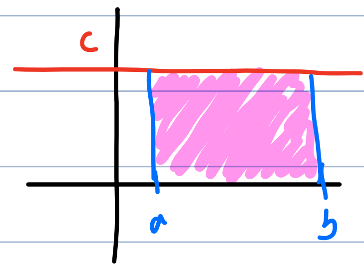
Integral properties

- reversing the limits of integration

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$$

idea: Δx becomes negative if you switch a and b

- $$\int_a^b c\,dx = \underbrace{(b-a)}_{\text{length}} \cdot \underbrace{c}_{\text{height}}$$

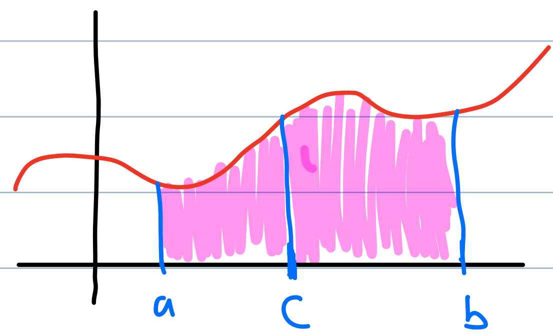


- $$\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

- $$\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$$

- $$\int_a^b [f(x)-g(x)]\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

- $$\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$$

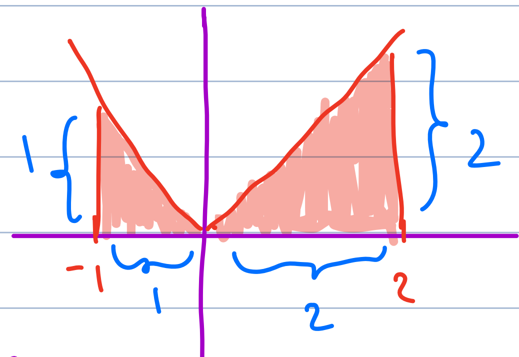


Example Evaluate $\displaystyle\int_{-1}^{2} (3|x| + 5)\,dx$

$$= \int_{-1}^{2} 3|x|\,dx + \underbrace{\int_{-1}^{2} 5\,dx}_{(2-(-1))\cdot 5 \,=\, 3\cdot 5 \,=\, 15}$$

$$= 3\underbrace{\int_{-1}^{2} |x|\,dx}_{?} + 15$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x \leq 0 \end{cases}$$



$$\int_{-1}^{2} |x|\,dx = \underbrace{\int_{-1}^{0} |x|\,dx}_{\frac{1\cdot 1}{2} = \frac{1}{2}} + \underbrace{\int_{0}^{2} |x|\,dx}_{\frac{2\cdot 2}{2} = 2} = \frac{1}{2} + 2 = \boxed{\frac{5}{2}}$$

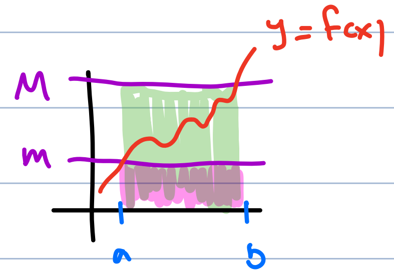
$$= 3\cdot\frac{5}{2} + 15 = \frac{15}{2} + 15 = \boxed{\frac{45}{2}}$$

Comparison properties

- if $f(x) \ge 0$ over $[a,b]$, then $\int_a^b f(x)\,dx \ge 0$

- if $f(x) \ge g(x)$ over $[a,b]$, then $\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$

- if $m \le f(x) \le M$ over $[a,b]$, then $\underbrace{\int_a^b m\,dx}_{} \le \int_a^b f(x)\,dx \le \underbrace{\int_a^b M\,dx}_{}$



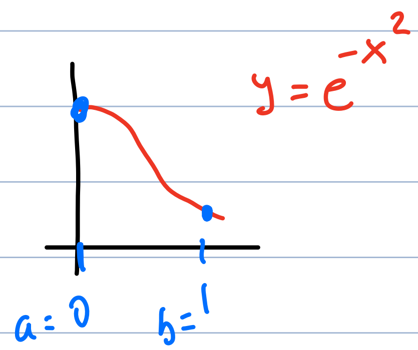
$$(b-a)\,m \le \int_a^b f(x)\,dx \le (b-a)\,M$$

Example Use comparison properties to give upper and lower bounds for $\int_0^1 e^{-x^2}\,dx$

First find an upper and lower bound for

$\underbrace{f(x) = e^{-x^2}}_{}$ over $[0,1]$

$\frac{1}{e^{x^2}}$ ... a decreasing function when $x \ge 0$



upper bound : $f(0) = \frac{1}{e^0} = 1 = M$

lower bound : $f(1) = \frac{1}{e^1} = \frac{1}{e} = m$

$$\underbrace{(b-a)}_{1}\,m \le \int_a^b f(x)\,dx \le \underbrace{(b-a)}_{1}\,M$$

$$\boxed{\frac{1}{e} \le \int_0^1 f(x)\,dx \le 1}$$

Evaluating integrals

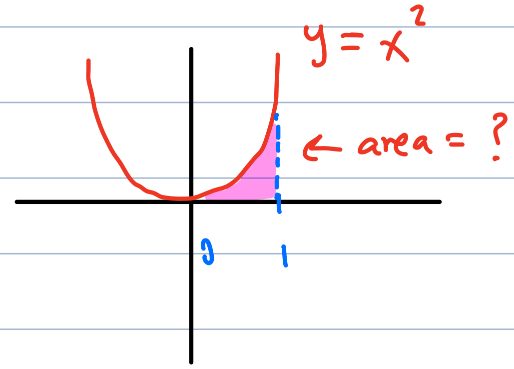
But how do we actually evaluate integrals?

We will see a hard way and an easier way (just like with derivatives)

Hard way: directly using the limit definition.

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\left(\sum_{i=1}^{n} f(x_i)\,\Delta x\right)$$

Example: find $\int_0^1 x^2\,dx$



$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\left(\sum_{i=1}^{n} f(x_i)\,\Delta x\right)$$

$a = 0,\quad b = 1$

$\Delta x = \dfrac{b-a}{n} = \dfrac{1-0}{n} = \dfrac{1}{n}$

$x_i = a + i\,\Delta x = 0 + i\cdot\left(\frac{1}{n}\right) = \dfrac{i}{n}$

$f(x) = x^2 \Rightarrow f(x_i) = x_i^2 = \left(\frac{i}{n}\right)^2$

$$\int_0^1 x^2\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} \underbrace{\left(\frac{i}{n}\right)^2}_{f(x_i)} \underbrace{\frac{1}{n}}_{\Delta x} = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{i^2}{n^3}$$

$$\frac{1^2}{n^3} + \frac{2^2}{n^3} + \frac{3^2}{n^3} + \cdots \frac{n^2}{n^3}$$

Useful sum formulas: $\bullet\ 1+2+3+4+\dots+n = \dfrac{n\cdot(n+1)}{2}$

$$\bullet\ 1^2+2^2+3^2+4^2+\dots+n^2 = \frac{n\cdot(n+1)(2n+1)}{6}$$

back to the problem:

$$\frac{1^2}{n^3}+\frac{2^2}{n^3}+\frac{3^2}{n^3}+\dots\frac{n^2}{n^3} = \frac{1}{n^3}\cdot\left(1^2+2^2+3^2+\dots+n^2\right)$$

$$= \frac{1}{n^3}\cdot\frac{n(n+1)(2n+1)}{6}$$

Taking the limit: $\displaystyle\lim_{n\to\infty}\left(\frac{n(n+1)(2n+1)}{6n^3}\right)$

$$= \lim_{n\to\infty}\left(\frac{n(n+1)(2n+1)}{6\,n\cdot n\cdot n}\right)$$

$$\frac{1}{6}\lim_{n\to\infty}\left(\frac{n}{n}\ \frac{(n+1)}{n}\ \frac{(2n+1)}{n}\right) = \frac{1}{6}\cdot 1\cdot 1\cdot 2 = \boxed{\frac{1}{3}}$$

$$\frac{n}{n} = 1,\qquad \frac{(n+1)}{n} = 1+\frac{1}{n} \to 1,\qquad \frac{(2n+1)}{n} = 2+\frac{1}{n} \to 2$$

This is tedious and difficut... is there an easier way?